DPCN 2026 · GROUP 24 · STATISTICS REPORT

Research output of three CSIR chemistry laboratories, 2016–2026

CSIR-NCL (Pune) · CSIR-IICT (Hyderabad) · CSIR-CECRI (Karaikudi)

14,352
distinct peer-reviewed papers

350,920
citations (OpenAlex, 13 Oct 2026)

63.3%
of ranked journal papers in Q1 (SJR)

347
roster scientists with ≥1 paper (of 505 listed)

Data: OpenAlex works affiliated to the three ROR-verified institutes (collected 13 Oct 2026; links with other Indian institutions 20 Oct 2026). 2026 is a partial year.
Quartiles: SCImago Journal Rank (best quartile of the publication year). Faculty: official institute staff pages.

Interactive dashboard: https://prathm01s.github.io/dpcn_almost_stable/

Data and method

14,352 distinct papers (2016 – 13 Oct 2026) from OpenAlex, one row per paper × institute author

Institute	Papers	Journal	Conference	Author rows	Roster scientists	With ≥1 paper	Quartile known
CSIR-IICT	6,167	6,086	81	20,183	287	169	94.1%
CSIR-NCL	5,404	5,353	51	14,726	167	152	95.3%
CSIR-CECRI	2,885	2,830	55	7,484	51	26	97.5%
All three (distinct)	14,352	14,166	186	42,393	505	347	95.2%

How the data was built

- Works pulled from OpenAlex by the institutes' ROR-verified IDs, with recall checks on CSIR/AcSIR affiliations
- Kept: articles, reviews, letters, conference papers; editorials, errata, preprints and book chapters excluded
- Faculty = scientific cadre from official staff pages, matched to OpenAlex authors seen in the corpus
- Quartile = SJR best quartile of the publication year (2025 values for 2026 papers)

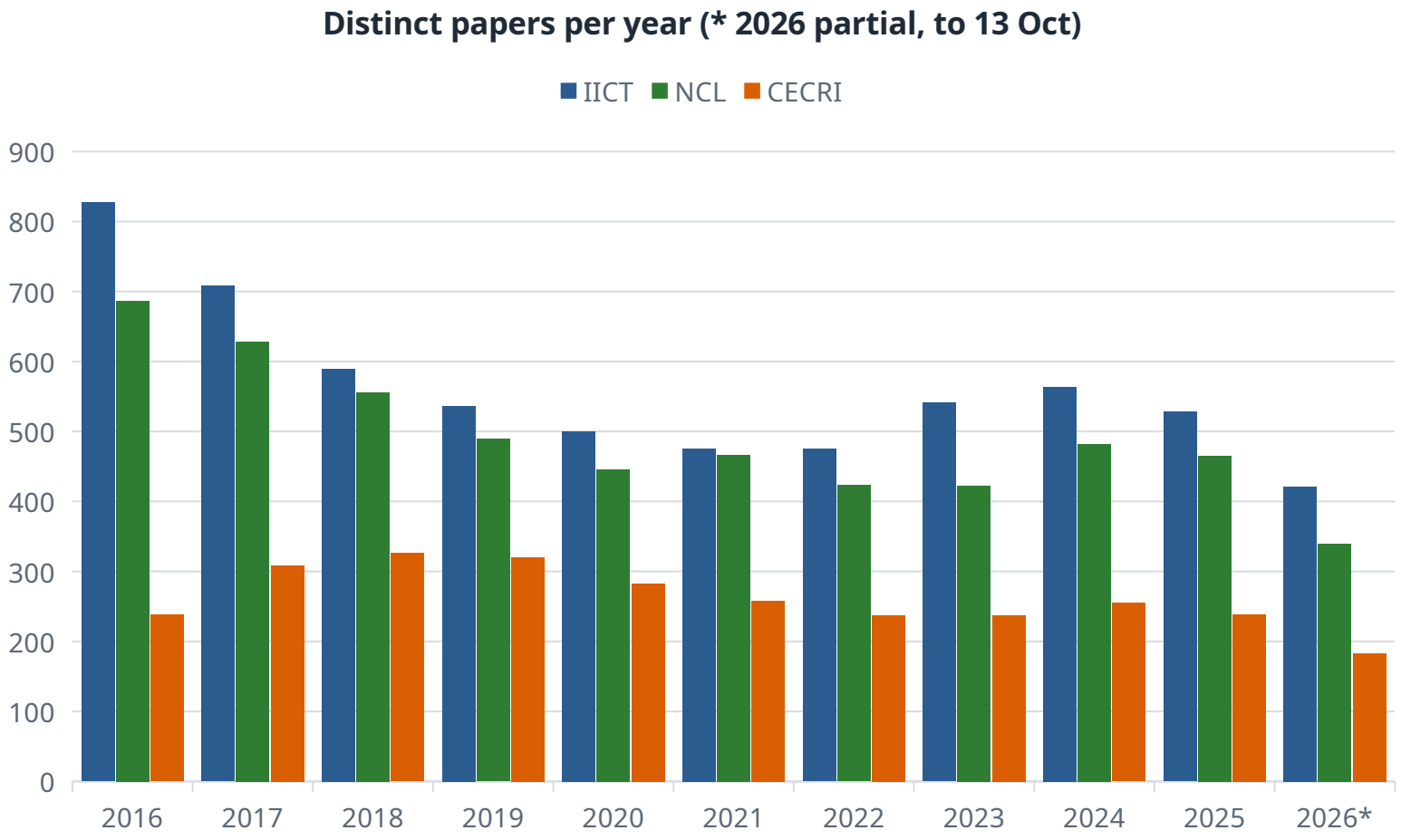
Limits to keep in mind

- OpenAlex author and affiliation matching is imperfect; some papers may be missing or mis-attributed
- h-index is computed on 2016–2026 corpus papers only, not career-wide; students/project staff are not faculty
- SJR quartiles can differ from JCR; citations are counts on the collection date
- None of the 186 conference papers has a CORE rank (CORE ranks computer-science venues)

Source: data/final/group24_data.csv, faculty_master.csv; stats/tables/quartile_distribution.csv, conference_ranks.csv

Publication trend

14,352 distinct papers in 2016–2026; IICT 43% of them, NCL 38%, CECRI 20%



Totals and growth

IICT: 6,167 papers; CAGR 2016→2025 −4.9%; peak year 2016

NCL: 5,404 papers; CAGR 2016→2025 −4.2%; peak year 2016

CECRI: 2,885 papers; CAGR 2016→2025 +0.1%; peak year 2018

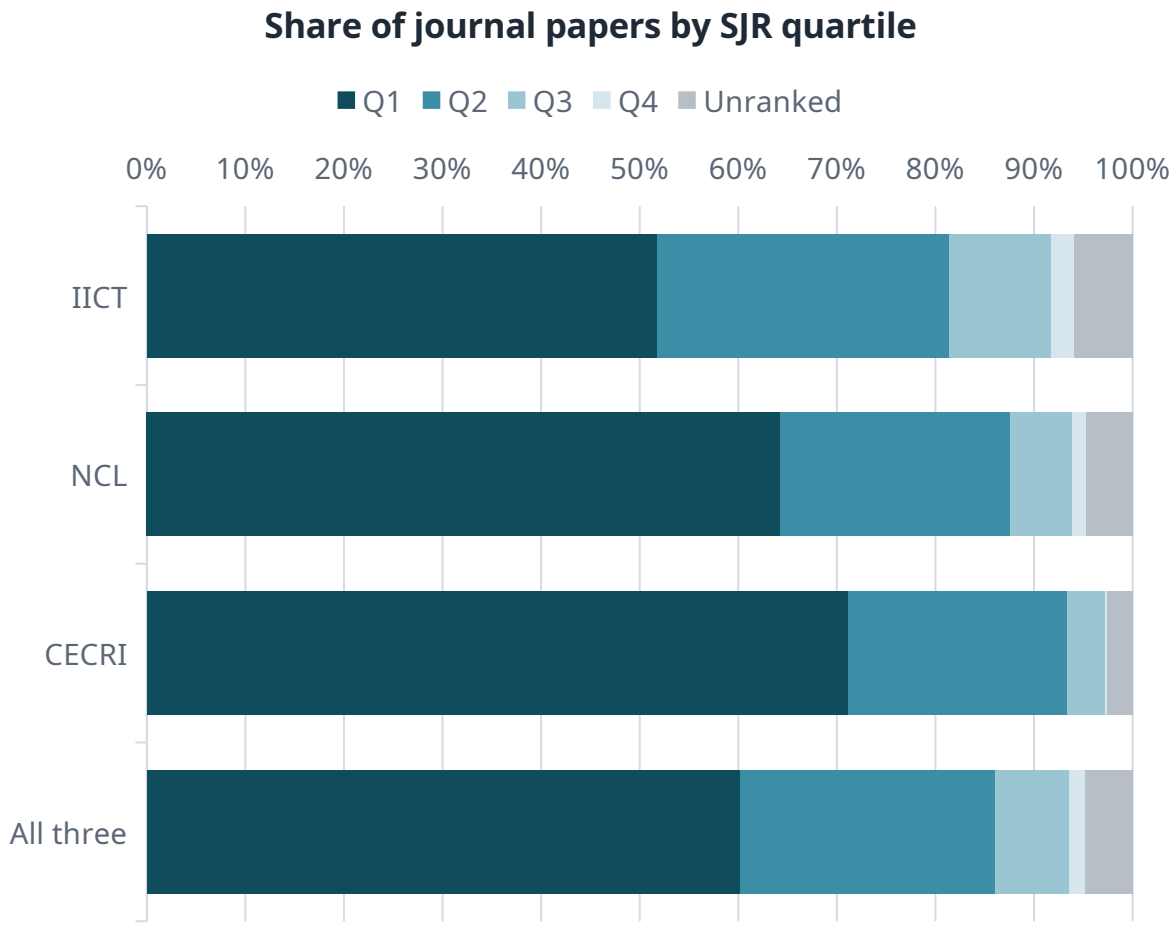
Observations

- Combined output went from 1,739 papers in 2016 to 1,220 in 2025
- Growth rates differ in sign across the three labs
- 2026 covers papers to 13 Oct only (934 so far) and is left out of CAGR

Source: stats/tables/publication_trend.csv · Papers counted once per institute; combined counts distinct papers

Journal quartiles and citation impact ratio

63.3% of ranked journal papers are in SJR-Q1 journals; Q1 by impact ratio (IR > 0.59, top 25% of papers) ranges from 22.0% (IICT) to 34.0% (CECRI)



Q1 by journal vs Q1 by impact ratio (% of papers)

Institute	Q1 by journal (SJR)	Q1 by IR (IR > 0.59)	Institute IR
IICT	55.1%	22.0%	0.41
NCL	67.4%	24.1%	0.47
CECRI	73.1%	34.0%	0.59
All three	63.3%	25.2%	0.47

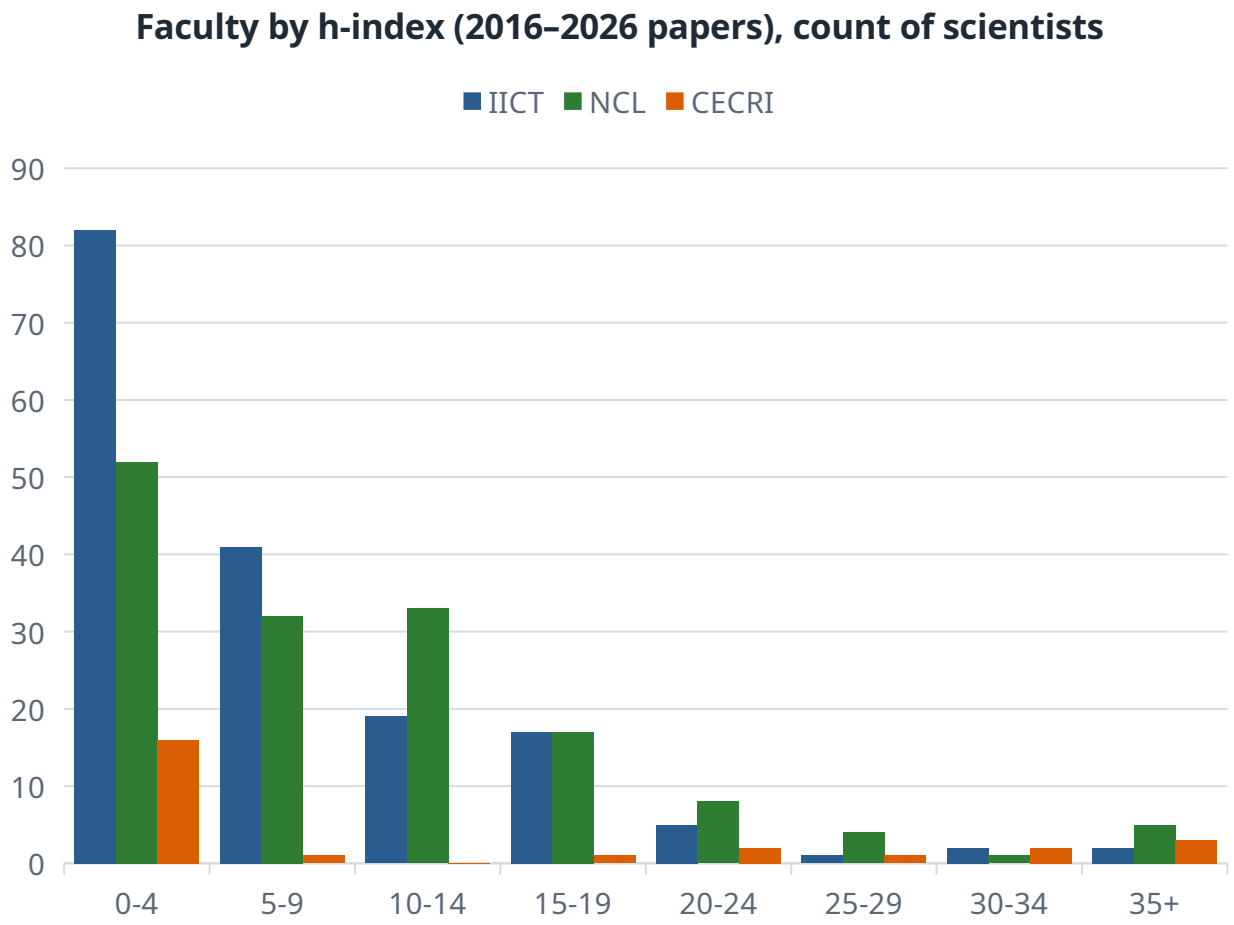
Impact ratio (IR) = citations received ÷ citations made (references). A paper is IR-Q1 when IR exceeds the threshold; the institute IR is all citations received ÷ all references made.

- Threshold IR > 0.59 is the 75th percentile of IR over all papers, so IR-Q1 is the top 25% of papers — the same share a quartile covers (median paper IR 0.26).
- The two definitions disagree: only 32.9% of papers in SJR-Q1 journals are top-25% by IR. SJR ranks the journal; IR ranks the paper.
- Older papers have had longer to be cited; 513 papers with no references in OpenAlex are excluded.

Source: stats/tables/quartile_distribution.csv, impact_ratio_summary.csv, impact_ratio_vs_sjr.csv · SJR best quartile (Scopus-based); IR from OpenAlex citations (13 Oct) and references

Faculty h-index distribution

Median h-index of the 347 roster scientists with papers is 6; CECRI 2.5, NCL 8, IICT 5



Institute	Faculty	Median	Mean	P75	Max
IICT	169	5	7.0	10	54
NCL	152	8	9.8	14	44
CECRI	26	2.5	11.9	23.5	59
All	347	6	8.6	13	59

Highest h-index (2016–2026 window)

Top faculty by h-index	Lab	Papers	Citations	h
Dr Subrata Kundu	CECRI	260	15,767	59
Dr. S. Venkata Mohan	IICT	230	11,389	54
Dr. Sreekumar Kurungot	NCL	181	8,978	44
Dr. Subashchandrabose Chi…	NCL	57	3,617	41
Dr. Ujjwal Pal	IICT	148	4,758	40
Dr. Vinod C Prabhakaran	NCL	152	4,752	39

Source: stats/tables/h_index_summary.csv, h_index_distribution.csv, faculty_performance.csv · h-index on 2016–2026 corpus papers

Top 5 journals per institute

Most frequent venues: IICT – ChemistrySelect (239); NCL – ChemistrySelect (140); CECRI – Electrochimica Acta (77)

CSIR-IICT 6,167 papers

#	Journal	Papers	Share	Q
1	ChemistrySelect	239	3.9%	Q2
2	Organic & Biomolecular Chemistry	144	2.4%	Q1
3	Tetrahedron Letters	128	2.1%	Q2
4	ChemInform	122	2.0%	—
5	The Journal of Organic Chemistry	120	2.0%	Q1

CSIR-NCL 5,404 papers

#	Journal	Papers	Share	Q
1	ChemistrySelect	140	2.6%	Q2
2	Chemical Communications	116	2.2%	Q1
3	The Journal of Organic Chemistry	106	2.0%	Q1
4	ACS Omega	97	1.8%	Q1
5	Organic Letters	94	1.8%	Q1

CSIR-CECRI 2,885 papers

#	Journal	Papers	Share	Q
1	Electrochimica Acta	77	2.7%	Q1
2	Journal of Materials Chemistry A	68	2.4%	Q1
3	ChemistrySelect	67	2.4%	Q2
4	New Journal of Chemistry	64	2.3%	Q1
5	International Journal of Hydrogen...	59	2.1%	Q1

All three combined: ChemistrySelect (443); New Journal of Chemistry (270); RSC Advances (245); Chemical Communications (239); The Journal of Organic Chemistry (225)

Source: stats/tables/top_journals.csv · ranked by distinct papers; share = % of the institute's papers; Q = SJR quartile

Top 10 most-cited papers (all three institutes)

350,920 citations in total; mean 24.4 and median 11 per paper; the top paper has 2,493

#	Title	Journal	Year	Lab	Citations	DOI
1	Recent Trends and Perspectives in Electrochemical Water Splitting with an Emphasis on Sulfi…	ACS Catalysis	2016	CECRI	2,493	10.1021/acscatal.6b02479
2	Cancer chemotherapy and beyond: Current status, drug candidates, associated risks and progr…	Genes & Diseases	2022	IICT	1,826	10.1016/j.gendis.2022.02.007
3	Precision and correctness in the evaluation of electrocatalytic water splitting: revisiting…	Energy & Environmental Science	2018	CECRI	1,623	10.1039/c7ee03457a
4	Covalent Organic Frameworks: Chemistry beyond the Structure	Journal of the American Chemi…	2018	NCL	1,574	10.1021/jacs.8b10334
5	Phytochemicals in Cancer Treatment: From Preclinical Studies to Clinical Practice	Frontiers in Pharmacology	2020	NCL	1,085	10.3389/fphar.2019.01614
6	Selective Molecular Separation by Interfacially Crystallized Covalent Organic Framework Thi…	Journal of the American Chemi…	2017	NCL	1,060	10.1021/jacs.7b06640
7	Cooperativity in Noncovalent Interactions	Chemical Reviews	2016	IICT	932	10.1021/cr500344e
8	“The Fe Effect”: A review unveiling the critical roles of Fe in enhancing OER activity of N…	Nano Energy	2020	CECRI	931	10.1016/j.nanoen.2020.105514
9	Functional Naphthalene Diimides: Synthesis, Properties, and Applications	Chemical Reviews	2016	IICT	927	10.1021/acs.chemrev.6b00160
10	Polylactic acid: synthesis and biomedical applications	Journal of Applied Microbiolo…	2019	NCL	873	10.1111/jam.14290

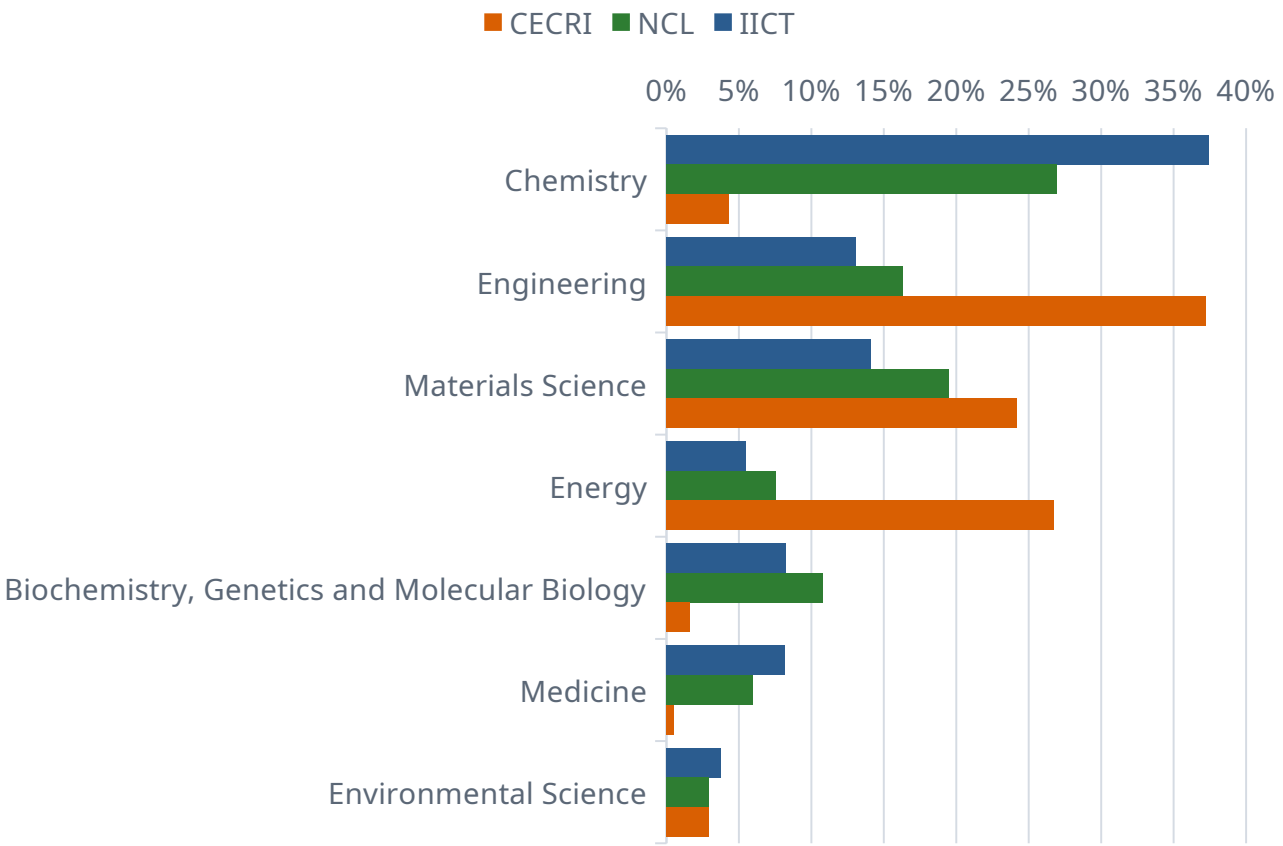
IICT: 125,231 citations, mean 20.3, uncited 12.2% · NCL: 134,679 citations, mean 24.9, uncited 11.0% · CECRI: 93,125 citations, mean 32.3, uncited 6.8%

Source: stats/tables/top_cited_papers.csv, citation_summary.csv · OpenAlex cited_by_count on 13 Oct 2026; older papers have had longer to accrue citations

Research areas and departments

Largest field: IICT – Chemistry (37%), NCL – Chemistry (27%), CECRI – Engineering (37%)

Share of each institute's papers by OpenAlex field



Top 2 departments per institute by papers

Department	Lab	Faculty	Papers	Q1
Applied Biology	IICT	21	509	56%
Polymers & Functional Materials	IICT	19	504	68%
Physical and Materials Chemistry	NCL	26	1,195	73%
Organic Chemistry	NCL	37	764	59%
Electrochemical Process Engineering	CECRI	6	350	89%
Electrodics and Electrocatalysis	CECRI	2	142	75%

Department = roster division of each faculty member; a paper counts in every department of its faculty authors. Papers by non-roster authors only are not assigned to a department.

Source: stats/tables/research_areas_field.csv, department_performance.csv · field = OpenAlex primary topic → field

Citation & Authorship Network Topology

27,268 citation links (1,653 cross-lab, in/out ratio 1.01); 7,864 authors in 130 Louvain communities (525 cross-lab links)

10,988
corpus papers in citation graph (27,268 in-cluster links, 1,653 cross-lab)

1.01
in / out citation ratio with other Indian insts (88,639 in, 87,292 out)

7,864
authors (35,423 co-author pairs, mean degree 9.0)

130
Louvain communities (modularity 0.81, 525 cross-lab links)

Directed Citation Network: Top PageRank Papers

#	PageRank top papers	Lab	Year	In-deg	PR
1	Recent Trends and Perspectives in Elect...	CECRI	2016	129	0.0030
2	Pt Nanoparticle Anchored Molecular Self...	CECRI	2016	41	0.0028
3	Unprotected and interconnected Ru 0 nan...	CECRI	2016	30	0.0027
4	Functional Naphthalene Diimides: Synthe...	IICT	2016	53	0.0020
5	Precision and correctness in the evalua...	CECRI	2018	110	0.0018
6	Waste biorefinery models towards sustai...	IICT	2016	71	0.0017

Undirected Authorship Network: Top Degree Collaborators

#	Most collaborative	Lab	Degree	Betweenness	Btw rank	Comm.
1	Balasubramanian Sridhar	IICT	380	0.096	#1	6
2	Dr. Jagadeesh Babu Nanub...	IICT	285	0.090	#2	18
3	Dr. Rajesh G. Gonnade	NCL	280	0.061	#5	4
4	Dr. Sunil Misra	IICT	198	0.055	#7	8
5	Dr. B.V. Subba Reddy	IICT	196	0.057	#6	6
6	Dr. Kumar Vanka	NCL	195	0.044	#9	4

Citation Flow with Other Indian Institutions

Lab	Cited by Indian	Cites Indian	In/out	Citing insts
IICT	32,840	38,891	0.84	1,816
NCL	32,434	30,206	1.07	1,683
CECRI	23,365	18,195	1.28	1,310

Top Brokers by Betweenness & Indian Partners

Balasubramanian Sridhar (IICT) · btw 0.096 · degree 380
Dr. Jagadeesh Babu Nanu... (IICT) · btw 0.090 · degree 285
Dr Subrata Kundu (CECRI) · btw 0.072 · degree 137
Top Indian co-authorship partners: Academy of Scientific and Innovative Rese... (641); Savitribai Phule Pune University (375); Council of Scientific and Industrial Rese... (337)

Source: stats/tables/network_summary.csv, network_citation_*.csv, network_authorship_*.csv, citation_flow_summary.csv · interactive graphs: https://prathm01s.github.io/dpcn_almost_stable/

Ten Non-Obvious Bibliometric & Network Findings

Key empirical findings from cluster network fragility, brain circulation, succession risk, and citation dynamics

#	Research Question	Empirical Finding & Headline Result
1	Do our researchers' co-authors have more co-authors than they do?	93.2% of researchers have fewer co-authors than their co-authors do. The average researcher has 9.2 co-authors, but the average co-author has 31.1.
2	Do the most-connected scientists collaborate mainly with each other?	The top 1% most-connected researchers show no rich-club effect and are less interconnected than chance (ρ_norm = 0.76). 41 of 78 are CSIR faculty.
3	How many researchers hold the cluster together?	Removing the top 872 bridge researchers (11.6% of the network) halves the cluster; random removal needs 3,279. Removing just 608 people disconnects the three labs from each other.
4	Is output concentrated in a few scientists?	√N = 19 scientists produce 36.2% of the cluster's papers (Price's law predicts 50%). The top 10% produce 50.3%; Gini = 0.63 (papers), 0.75 (citations).
5	Is a Q1 journal a reliable signal of a highly cited paper?	11.3% of the cluster's top-10% most-cited papers were published outside Q1 journals, while 1.4% of Q1 papers are still uncited.
6	Are the papers most influential inside our cluster the same as the world's most-cited?	In-cluster influence and global citations correlate only moderately (Spearman ρ = 0.40). 82 papers are 'hidden gems': top-5% internal influence, below-median global citations.
7	Which papers were ignored for years and then suddenly took off?	15 papers show delayed recognition. The strongest, 'Novel triazole hybrids of myrrhanone C, a natural polypodane triterpene: Synthesis, cytotoxic activity and cell based studies' (CSIR-IICT, 2016), averaged 3.6 citations/yr for 5 years, then jumped to 6.3/yr from 2021.
8	How much output did COVID cost, and did research shift?	Output was 1.8% above trend in 2020 and 12.7% above in 2021. COVID-related work peaked at 5.5% of CSIR-IICT's papers in 2021; the topic mix shifted 1.05× more than in a normal period.
9	How many of our scientists trained in India, worked abroad, and came back, and did time abroad change their research?	8.3% of classifiable CSIR scientists are returnees (NCL 13.2%, IICT 4.0%, CECRI 0.0%), mostly via US, DE. Returnees' papers show no significant difference in citations than home-grown peers of similar career age, and 9.0% vs 16.3% of their papers have foreign co-authors.
10	How much of each lab's output depends on scientists close to retirement?	20.0% of CSIR-IICT's recent papers (23.3% of its citations) are led solely by scientists likely to retire within ~5 years. Highest exposure: Energy & Environmental Engg. (69.3%).

Source: stats/tables/insights/*.json · interactive charts and methodology notes: https://prathm01s.github.io/dpcn_almost_stable/